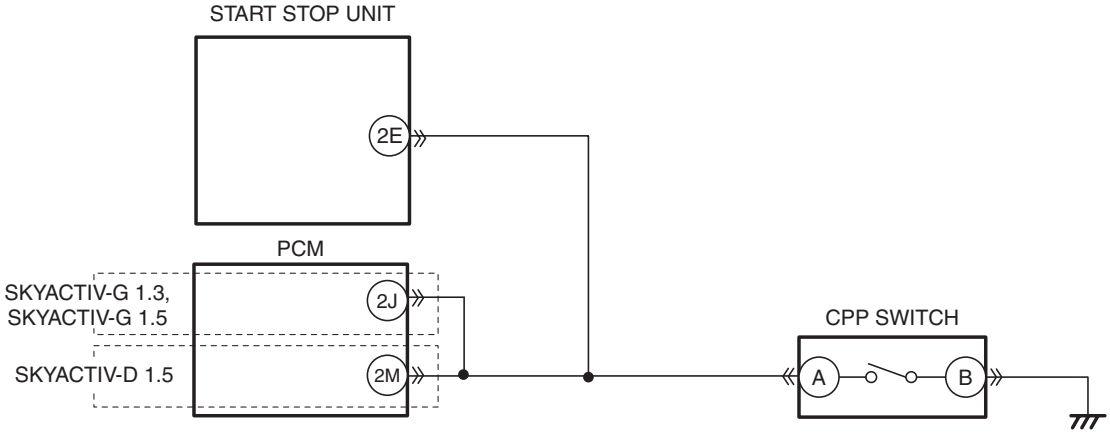
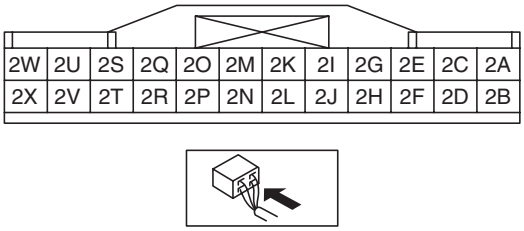
DTC P0830:23 [START STOP UNIT]

id0902p6023100

Description	CPP switch circuit malfunction
Detection condition	• Start stop unit detects that the CPP switch is stuck on for **10 min or more**.
Fail-safe	• Change to the back-up mode. • Perform the control with CPP switch turned off.
Possible cause	• DTCs are stored in the PCM. • CPP switch connector or terminal malfunction • CPP switch malfunction • Start stop unit connector or terminal malfunction • PCM connector or terminal malfunction • Short to ground in the wiring harness between the following terminals: — CPP switch terminal A—PCM terminal 2J (SKYACTIV-G 1.3, SKYACTIV-G 1.5) — CPP switch terminal A—PCM terminal 2M (SKYACTIV-D 1.5) — CPP switch terminal A—Start stop unit terminal 2E • Start stop unit malfunction



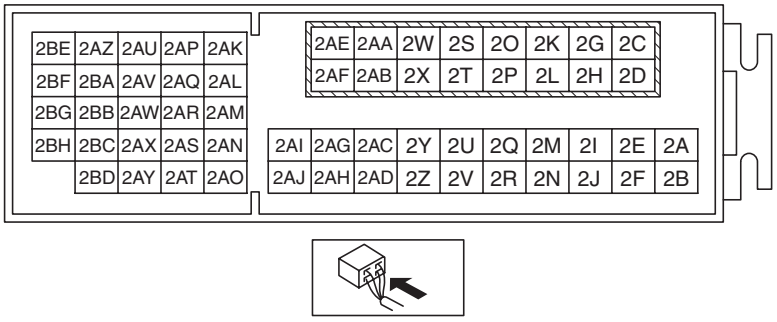
START STOP UNIT
WIRING HARNESS-SIDE CONNECTOR



CPP SWITCH
WIRING HARNESS-SIDE CONNECTOR



PCM
WIRING HARNESS-SIDE CONNECTOR



Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY PCM DTCs** • Retrieve the PCM DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
2	**INSPECT CPP SWITCH CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the CPP switch connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 7.
3	**INSPECT CPP SWITCH** • Inspect the CPP switch. (See CLUTCH PEDAL POSITION (CPP) SWITCH INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See CLUTCH PEDAL POSITION (CPP) SWITCH INSPECTION [SKYACTIV-D 1.5].) • Is the CPP switch normal?	Yes	Go to the next step.
		No	Replace the CPP switch, then go to the Step 7. (See CLUTCH PEDAL POSITION SWITCH REMOVAL/ INSTALLATION [F65M-R].) (See CLUTCH PEDAL POSITION SWITCH REMOVAL/ INSTALLATION [F66M-R].)
4	**INSPECT START STOP UNIT CONNECTOR CONDITION** • Disconnect the start stop unit connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 7.
5	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 7.
6	**INSPECT CPP SWITCH CIRCUIT FOR SHORT TO GROUND** • Verify that the CPP switch, start stop unit and PCM connectors are disconnected. • Inspect for continuity between CPP switch terminal A (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • CPP switch terminal A—PCM terminal 2J (SKYACTIV-G 1.3, SKYACTIV-G 1.5) • CPP switch terminal A—PCM terminal 2M (SKYACTIV-D 1.5) • CPP switch terminal A—Start stop unit terminal 2E **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to the next step.
		No	Go to the next step.

Step	Inspection		Action
7	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the start stop unit using the M-MDS. (See CLEARING DTC [START STOP UNIT].) • Switch the ignition ON (engine off or on) and wait for **10 min or more**. • Retrieve the start stop unit DTCs using the M-MDS. (See DTC INSPECTION [START STOP UNIT].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the start stop unit. (See START STOP UNIT REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
8	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [START STOP UNIT].)
		No	DTC troubleshooting completed.